



**Marwadi University**  
**Faculty of Technology**  
**Department of Information and Communication Technology**

**Digital Signal and Image  
Processing (01CT1513)**

**Aim : Generate Histogram of Images, apply Histogram Equalization**

**Experiment : 8**

**Date: 014/08/2026**

**Enrollment No: 92400133125**

**Aim : Generate Histogram of Images, apply Histogram Equalization**

**Code :**

```
import os
import cv2
import numpy as np
import matplotlib.pyplot as plt

script_dir = os.path.dirname(os.path.abspath(__file__))
image_path = os.path.join(script_dir, "4.jpg")
reference_path = os.path.join(script_dir, "3.jpg")

image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Image not found")
else:
    histogram = cv2.calcHist([image], [0], None, [256], [0, 256])

    plt.figure(figsize=(8, 6))
    plt.title("Original Histogram")
    plt.xlabel("Pixel Value")
    plt.ylabel("Frequency")
    plt.plot(histogram)
    plt.xlim([0, 256])
    plt.grid(True)
    plt.show()

    equalized_image = cv2.equalizeHist(image)

    plt.figure(figsize=(10, 5))

    plt.subplot(1, 2, 1)
    plt.title("Original Image")
```

```
plt.imshow(image, cmap="gray")  
plt.axis("off")
```

```
plt.subplot(1, 2, 2)  
plt.title("Equalized Image")  
plt.imshow(equalized_image, cmap="gray")  
plt.axis("off")
```

```
plt.tight_layout()  
plt.show()
```

```
equalized_histogram = cv2.calcHist(  
    [equalized_image], [0], None, [256], [0, 256]  
)
```

```
plt.figure(figsize=(8, 6))  
plt.title("Equalized Histogram")  
plt.xlabel("Pixel Value")  
plt.ylabel("Frequency")  
plt.plot(equalized_histogram)  
plt.xlim([0, 256])  
plt.grid(True)  
plt.show()
```

```
reference = cv2.imread(reference_path, cv2.IMREAD_GRAYSCALE)
```

```
if reference is None:
```

```
    print("Reference image not found")
```

```
else:
```

```
    source_hist = cv2.calcHist(  
        [image], [0], None, [256], [0, 256]  
    )
```

```
    reference_hist = cv2.calcHist(  
        [reference], [0], None, [256], [0, 256]  
    )
```



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```
source_hist = source_hist / source_hist.sum()
reference_hist = reference_hist / reference_hist.sum()
```

```
source_cdf = source_hist.cumsum()
reference_cdf = reference_hist.cumsum()
```

```
mapping = np.interp(
    source_cdf,
    reference_cdf,
    np.arange(256)
)
```

```
matched_image = mapping[image].astype(np.uint8)
```

```
plt.figure(figsize=(15, 5))
```

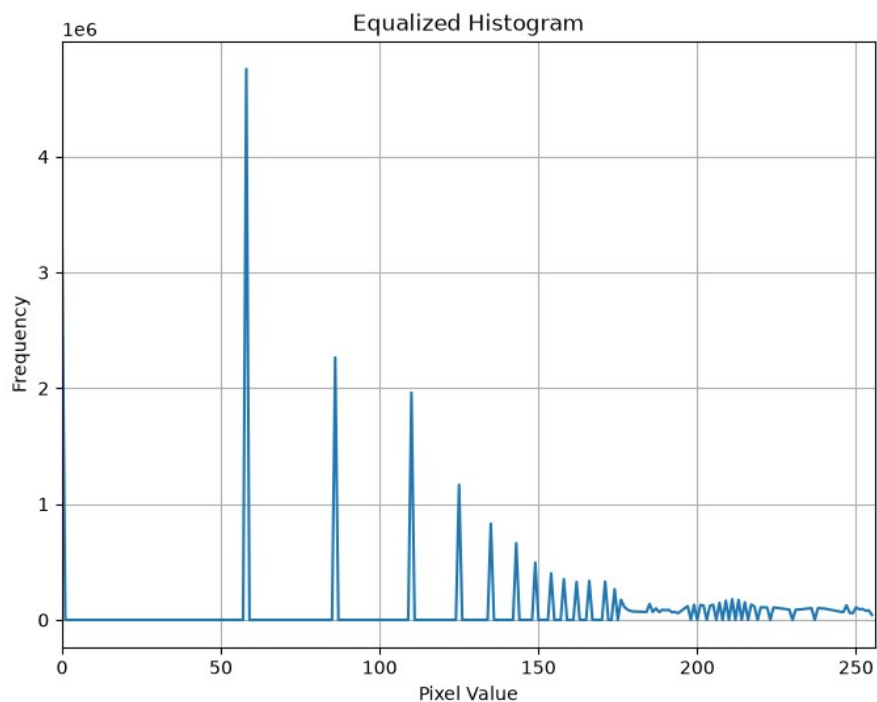
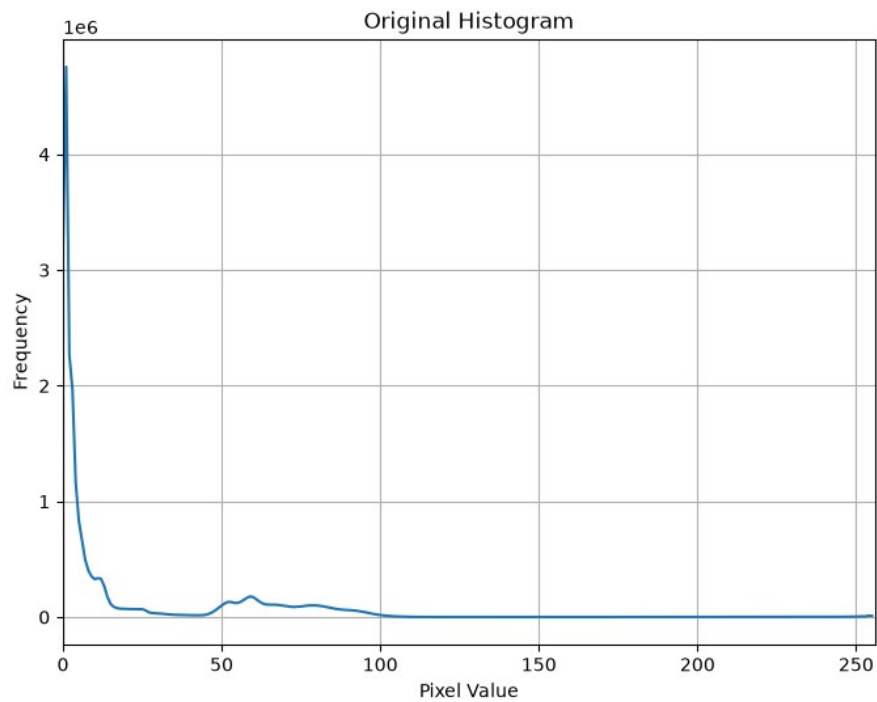
```
plt.subplot(1, 3, 1)
plt.title("Source Image")
plt.imshow(image, cmap="gray")
plt.axis("off")
```

```
plt.subplot(1, 3, 2)
plt.title("Reference Image")
plt.imshow(reference, cmap="gray")
plt.axis("off")
```

```
plt.subplot(1, 3, 3)
plt.title("Matched Image")
plt.imshow(matched_image, cmap="gray")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

Output :



Original Image



Equalized Image





|                                                                                                            |                                                                                                                            |                                   |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|  <b>Marwadi University</b> | <b>Marwadi University</b><br><b>Faculty of Technology</b><br><b>Department of Information and Communication Technology</b> |                                   |
| <b>Digital Signal and Image Processing (01CT1513)</b>                                                      | <b>Aim : Generate Histogram of Images, apply Histogram Equalization</b>                                                    |                                   |
| <b>Experiment : 8</b>                                                                                      | <b>Date: 014/08/2026</b>                                                                                                   | <b>Enrollment No: 92400133125</b> |



Original :



|                                                                                                               |                                                                                                                            |                                   |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
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| <b>Digital Signal and Image Processing (01CT1513)</b>                                                         | <b>Aim : Generate Histogram of Images, apply Histogram Equalization</b>                                                    |                                   |
| <b>Experiment : 8</b>                                                                                         | <b>Date: 014/08/2026</b>                                                                                                   | <b>Enrollment No: 92400133125</b> |

Refrance Image :



Conclusion :

This experiment helped us understand how histogram processing can be used to improve the quality and contrast of a grayscale image. Histogram equalization enhanced the image by spreading the pixel intensity values, making important details more visible. Histogram matching was also performed by using a reference image to adjust the intensity distribution of the source image. Overall, the experiment gave a practical understanding of how histograms can be used for image enhancement and improving visual appearance.